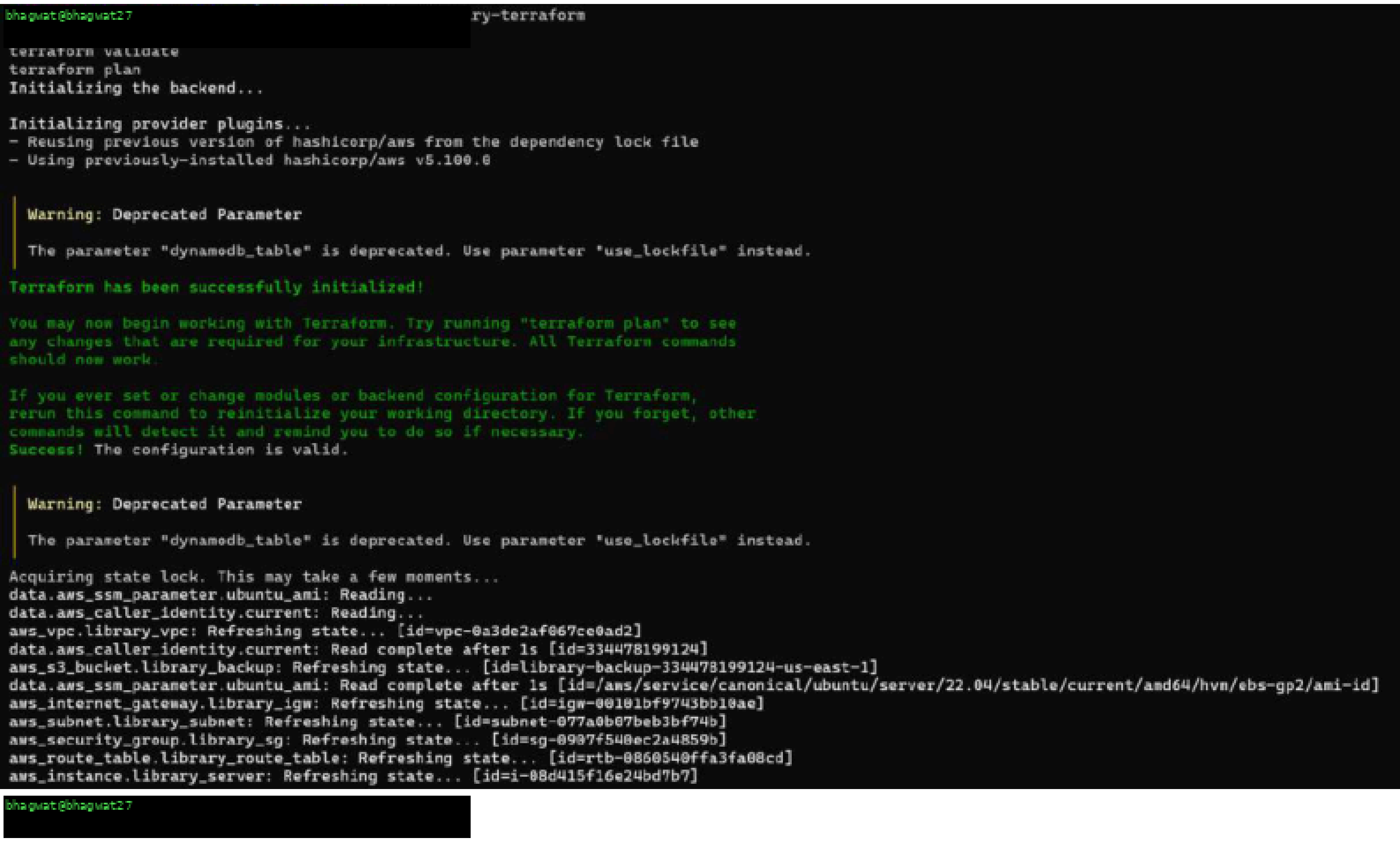
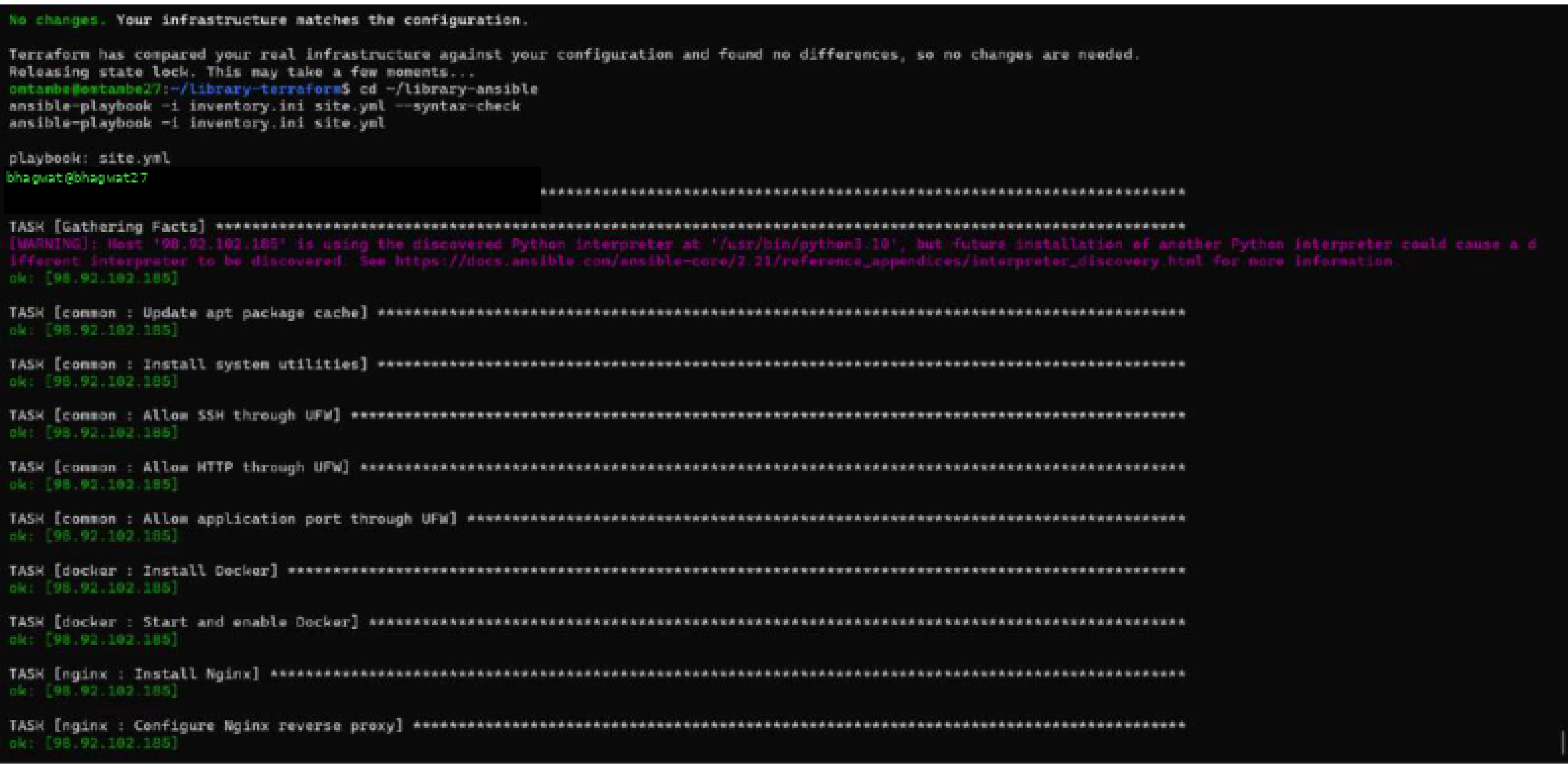
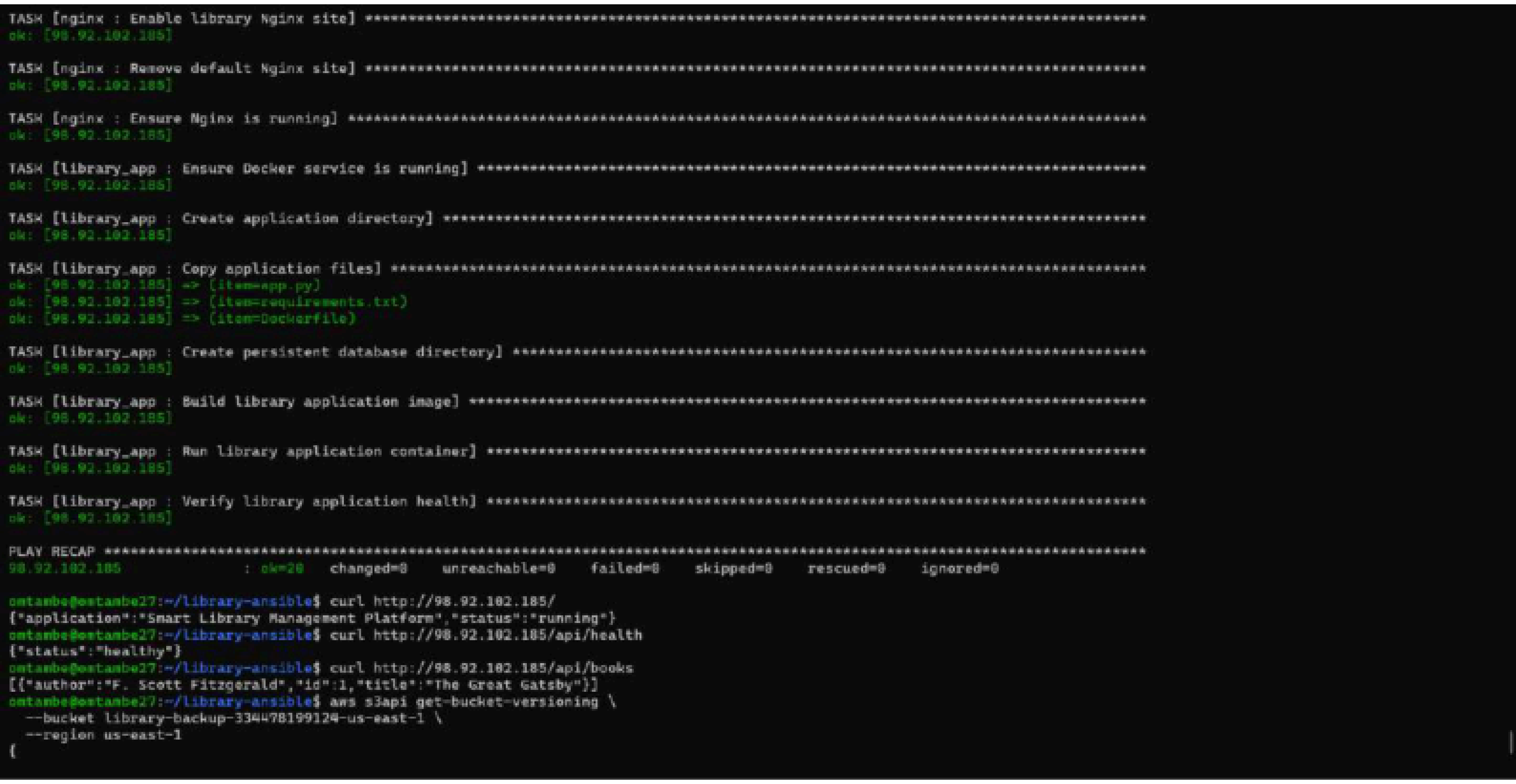


```
bhagwat@bhagwat27 [redacted] ry-terraform
terraform validate
terraform plan
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0


│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.


│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│
Acquiring state lock. This may take a few moments...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_caller_identity.current: Reading...
aws_vpc.library_vpc: Refreshing state... [id=vpc-0a3de2af067ce0ad2]
data.aws_caller_identity.current: Read complete after 1s [id=334478199124]
aws_s3_bucket.library_backup: Refreshing state... [id=library-backup-334478199124-us-east-1]
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
aws_internet_gateway.library_igw: Refreshing state... [id=igw-00101bf9743bb10ae]
aws_subnet.library_subnet: Refreshing state... [id=subnet-077a0b07beb3bf74b]
aws_security_group.library_sg: Refreshing state... [id=sg-0907f540ec2a4859b]
aws_route_table.library_route_table: Refreshing state... [id=rtb-0860540ffa3fa08cd]
aws_instance.library_server: Refreshing state... [id=i-08d415f16e24bd7b7]
bhagwat@bhagwat27 [redacted]
```

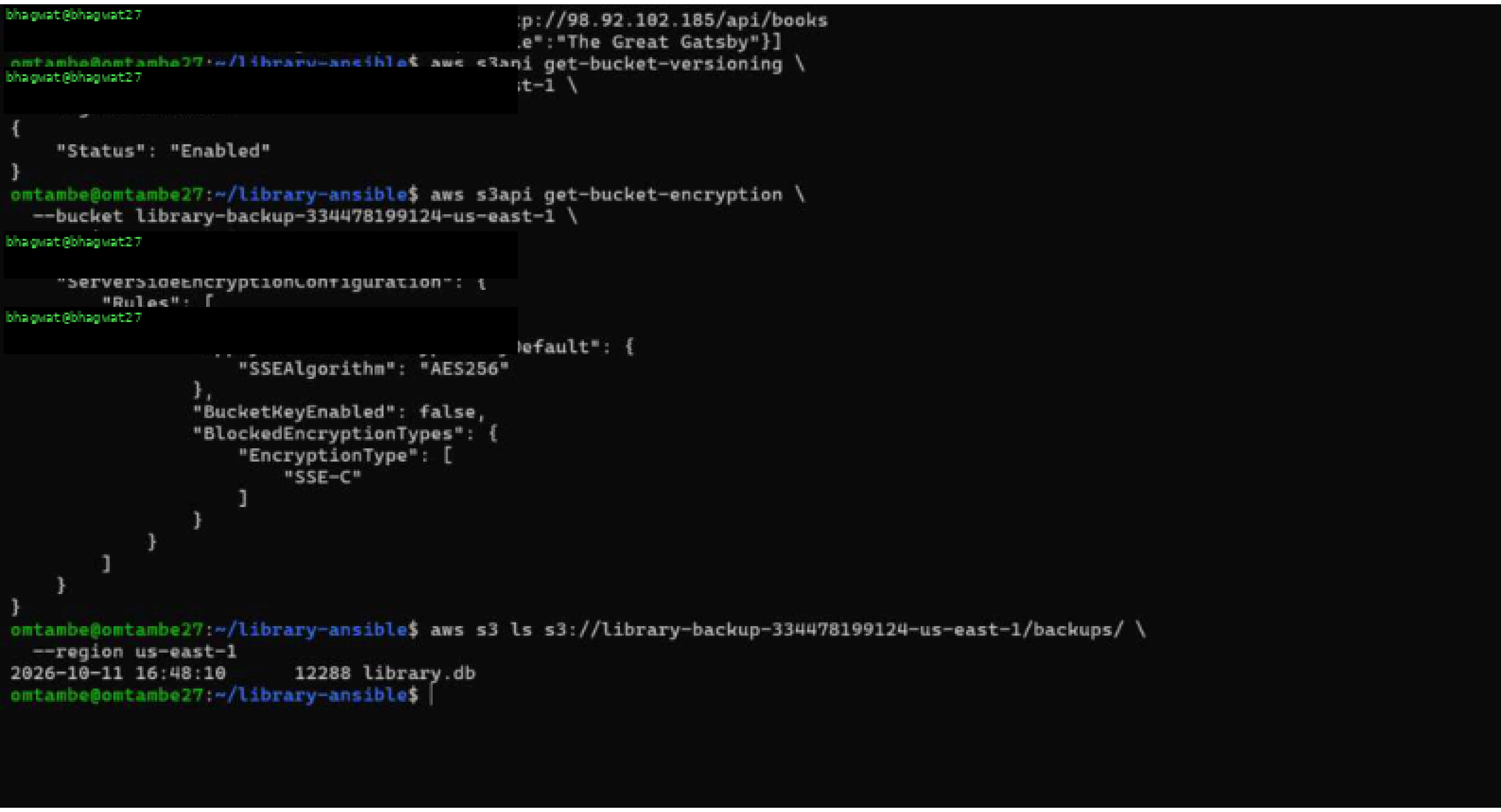


```
No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
omtambe@omtambe27:~/library-terraform$ cd ~/library-ansible
ansible-playbook -i inventory.ini site.yml --syntax-check
ansible-playbook -i inventory.ini site.yml

playbook: site.yml
bhagwat@bhagwat27
[redacted] ************************************************************************

TASK [Gathering Facts] ************************************************************************************************************
[WARNING]: Host '98.92.102.185' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a d
ifferent interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [98.92.102.185]

TASK [common : Update apt package cache] ******************************************************************************************
ok: [98.92.102.185]

TASK [common : Install system utilities] ******************************************************************************************
ok: [98.92.102.185]

TASK [common : Allow SSH through UFW] *********************************************************************************************
ok: [98.92.102.185]

TASK [common : Allow HTTP through UFW] ********************************************************************************************
ok: [98.92.102.185]

TASK [common : Allow application port through UFW] ********************************************************************************
ok: [98.92.102.185]

TASK [docker : Install Docker] ****************************************************************************************************
ok: [98.92.102.185]

TASK [docker : Start and enable Docker] *******************************************************************************************
ok: [98.92.102.185]

TASK [nginx : Install Nginx] ******************************************************************************************************
ok: [98.92.102.185]

TASK [nginx : Configure Nginx reverse proxy] **************************************************************************************
ok: [98.92.102.185]
```



```
TASK [nginx : Enable library Nginx site] *******************************************************************************************
ok: [98.92.102.185]

TASK [nginx : Remove default Nginx site] *******************************************************************************************
ok: [98.92.102.185]

TASK [nginx : Ensure Nginx is running] *********************************************************************************************
ok: [98.92.102.185]

TASK [library_app : Ensure Docker service is running] ******************************************************************************
ok: [98.92.102.185]

TASK [library_app : Create application directory] *********************************************************************************
ok: [98.92.102.185]

TASK [library_app : Copy application files] ****************************************************************************************
ok: [98.92.102.185] => (item=app.py)
ok: [98.92.102.185] => (item=requirements.txt)
ok: [98.92.102.185] => (item=Dockerfile)

TASK [library_app : Create persistent database directory] **************************************************************************
ok: [98.92.102.185]

TASK [library_app : Build library application image] *******************************************************************************
ok: [98.92.102.185]

TASK [library_app : Run library application container] *****************************************************************************
ok: [98.92.102.185]

TASK [library_app : Verify library application health] *****************************************************************************
ok: [98.92.102.185]

PLAY RECAP *************************************************************************************************************************
98.92.102.185              : ok=20   changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

omtambe@omtambe27:~/library-ansible$ curl http://98.92.102.185/
{"application":"Smart Library Management Platform","status":"running"}
omtambe@omtambe27:~/library-ansible$ curl http://98.92.102.185/api/health
{"status":"healthy"}
omtambe@omtambe27:~/library-ansible$ curl http://98.92.102.185/api/books
[{"author":"F. Scott Fitzgerald","id":1,"title":"The Great Gatsby"}]
omtambe@omtambe27:~/library-ansible$ aws s3api get-bucket-versioning \
  --bucket library-backup-334478199124-us-east-1 \
  --region us-east-1
{
```

```
bhagwat@bhagwat27
```



```
bhagwat@bhagwat27 [redacted] :p://98.92.102.185/api/books
                             .e":"The Great Gatsby"}]
omtambe@omtambe27:~/library-ansible$ aws s3api get-bucket-versioning \
bhagwat@bhagwat27 [redacted] :t-1 \
{
    "Status": "Enabled"
}
omtambe@omtambe27:~/library-ansible$ aws s3api get-bucket-encryption \
  --bucket library-backup-334478199124-us-east-1 \
bhagwat@bhagwat27 [redacted]
    "ServerSideEncryptionConfiguration": {
        "Rules": [
bhagwat@bhagwat27 [redacted] Default": {
                    "SSEAlgorithm": "AES256"
                },
                "BucketKeyEnabled": false,
                "BlockedEncryptionTypes": {
                    "EncryptionType": [
                        "SSE-C"
                    ]
                }
            }
        ]
    }
}
omtambe@omtambe27:~/library-ansible$ aws s3 ls s3://library-backup-334478199124-us-east-1/backups/ \
  --region us-east-1
2026-10-11 16:48:10      12288 library.db
omtambe@omtambe27:~/library-ansible$ |
```

bhagwat@bhagwat27 [redacted]